

AB POWER ENGINEERING

Engr. Alexis T. Belonio, MS, ABE, ASEAN Engineer
*Freelance Consultant and Technology Developer; Former Senior Research Fellow (PhilRice)
and Associate Professor (Central Philippine University)*
Member: PSABE, ASABE, ASME

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INTERNAL COMBUSTION ENGINE

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1. An engine that burns fuel inside the cylinder.

- a. External combustion engine
- b. Internal combustion engine
- c. Sterling engine
- d. None of the above

2. The part of an engine that slides up and down the cylinder and transmits power to the crankshaft.

- a. Connecting rod
- b. Cylinder block
- c. Piston
- d. Spark plug
- e. None of the above

3. The amount of power that the manufacturer specified for an engine.

- a. Indicated horsepower
- b. Brake horsepower
- c. Rated horsepower
- d. None of the above

4. A 100-pferde-starke rated engine is equal to ____ hp rating.

- a. 96.8
- b. 98.6
- c. 101.4
- d. None of the above

5. The mechanical breaking up of fuel into spray droplets by applying high-velocity air passing through a jet.

- a. Atomization
- b. Carburetion
- c. Turbo charging
- d. None of the above

6. The constant volume combustion cycle.

- a. Otto cycle
- b. Diesel cycle
- c. Dual cycle
- d. None of the above

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7. The constant pressure combustion cycle.

- a. Otto cycle
- b. Diesel cycle
- c. Dual cycle
- d. None of the above

8. The distance of the movement of piston from the top dead to the bottom dead center.

- a. Stroke
- b. Bore
- c. Valve clearance
- d. None of the above

9. The decrease or increase of engine speed or flow of fuel.

- a. Throttling
- b. Clutching
- c. Atomization
- d. None of the above

10. A device that uses fluid power to produce mechanical force and motion.

- a. Governor
- b. Accumulator
- c. Actuator
- d. None of the above

11. A device that is used to actuate the throttle of an engine to increase its speed.

- a. Governor
- b. Accelerator
- c. Actuator
- d. None of the above

12. A device that meters the fuel and the air in spark-ignition engines and mixes them into a harmonious mixture before the mixture enters the combustion chamber of an engine.

- a. Governor
- b. Air injector
- c. Carburetor
- d. All of the above

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13. The usable power delivered by the engine.

- a. Indicated horsepower
- ☒ b. Brake horsepower
- c. Rated horsepower
- d. None of the above

14. A device that permits the flow of fuel and of burned gases in one direction only.

- ☒ a. Check valve
- b. Relief valve
- c. Spark plug
- d. None of the above

15. An internal combustion engine that produces power in one revolution of the crankshaft.

- a. Four-stroke-cycle engine
- ☒ b. Two-stroke-cycle engine
- c. Rotary engine
- d. None of the above

16. The ratio of total volume to the clearance volume.

- ☒ a. Compression ratio
- b. Air-fuel ratio
- c. Bore stroke ratio
- d. None of the above

17. The volume displaced by the piston in making an upward or a downward stroke.

- ☒ a. Piston displacement volume
- b. Clearance volume
- c. Total volume
- d. None of the above

18. An engine has a piston displacement of 38.4 in³. If the clearance volume of the cylinder is 2.5 in³, what is its compression ratio?

- ☒ a. 1:12
- ☒ b. 1:16
- c. 1:8
- d. None of the above

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19. The ratio of the piston displacement volume plus the clearance volume over the clearance volume.

- ☒ a. Compression ratio
- b. Air-fuel ratio
- c. Bore-stroke ratio
- d. None of the above

20. Suppose a 4-cylinder engine has 3.5-in. bore and 4.0-in. stroke, what is the engine displacement.

- ☒ a. 154 in³
- b. 175 in³
- c. 225 in³
- d. None of the above

21. A device that provides ignition to a gasoline engine.

- a. Fuel injector
- b. Carburetor
- ☒ c. Spark plug
- d. None of the above

22. During the power stroke, which of the following valve is open?

- a. Intake
- b. Exhaust
- c. Intake and exhaust
- ☒ d. None of the above

23. The number of spark plugs for a 6-cylinder diesel engine.

- a. 6
- b. 12
- c. 4
- ☒ d. None of the above

24. At the downward stroke of the two-stroke cycle engine, which of the events are taking place?

- ☒ a. Power, intake, and exhaust
- b. Compression
- c. Power & exhaust
- d. None of the above

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25. An engine that injects fuel into the combustion chamber near the end of the combustion stroke and is subsequently ignited only by heat of compression.

- a. Gasoline engine
- ☒ b. Diesel engine
- c. Steam engine
- d. None of the above

26. The power transmitted to the piston by the gas in the cylinder.

- a. Brake horsepower
- ☒ b. Indicated horsepower
- c. Drawbar horsepower
- d. All of the above

27. The number of piston explosion in a single-cylinder, four-stroke cycle engine and in 100 revolution of the crankshaft.

- a. 100
- ☒ b. 50
- c. 25
- d. None of the above

28. A device that connects and disconnects the power input to and from the power output.

- ☒ a. Clutch
- b. Gear
- c. Pulley
- d. None of the above

29. The space when the piston is at the top dead center.

- a. Piston displacement volume
- ☒ b. Clearance volume
- c. Total volume
- e. None of the above

30. How many piston strokes are there in one revolution of the crankshaft?

- a. One
- ☒ b. Two
- c. Four
- d. None of the above

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31. The number of times the exhaust valve opens in a 2-stroke-cycle engine running at 2000 rpm.

- a. 1000
- b. 2000
- c. 1500
- ☒ d. None of the above

32. What is the indicated horsepower of a four-stroke, six-cylinder engine with 4" bore and 4" stroke. The engine is running at 1500 rpm and the mean effective pressure is 80 psi.

- a. 35 hp
- ☒ b. 45 hp
- c. 52 hp
- d. None of the above

33. The sequence by which the cylinder delivers the power stroke.

- a. Ignition
- ☒ b. Firing order
- c. Flash point
- d. None of the above

34. An engine that burns the fuel outside the cylinder.

- a. Spark-ignition engine
- ☒ b. External combustion engine
- c. Compression-ignition engine
- d. None of the above

35. What comes in during the intake stroke of a compression-ignition engine?

- a. Fuel and air
- b. Fuel
- ☒ c. Air
- d. None of the above

36. The firing order of a 4-cylinder engine.

- a. 1-2-3-4
- ☒ b. 1-3-4-2 / 1-2-4-3
- c. 1-4-2-3
- d. None of the above

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37. Two pistons running at the same time in the same direction but performing different strokes.

- a. Running mate
- b. Firing order
- c. Valve timing
- d. None of the above

38. How many times the intake valve opens in a 4-stroke-cycle engine running at 2000 rpm.

- a. 1000
- b. 2000
- c. 500
- d. None of the above

39. A four-stroke, six-cylinder biomass gas producer engine has 8-inches cylinder bore and 12-inches stroke. If the engine is running at 500 rpm speed, what is its indicated horsepower? The mean effective pressure is 80 psi.

- a. 22.5 hp
- b. 30.5 hp
- c. 40.5 hp
- d. None of the above

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40. The engine brake thermal efficiency.

- a. Brake hp over fuel hp
- b. Rated hp over indicated hp
- c. Friction hp over flywheel hp
- d. None of the above

41. The number of times the intake valve opens in a single-cylinder, four-stroke cycle engine after completing 200 revolutions.

- a. 200
- b. 100
- c. 50
- d. None of the above

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42. A device that creates ignition of fuel in a diesel engine.

- a. Spark plug
- b. Piston
- c. Carburetor
- d. None of the above

43. The part of an engine that converts reciprocating motion to rotary motion.

- a. Crankshaft
- b. Piston
- c. Camshaft
- d. None of the above

44. The firing order of a 6-cylinder V-piston straight engine.

- a. 1-5-3-6-2-4
- b. 1-4-2-6-3-5
- c. 1-2-3-4-5-6
- d. None of the above

45. The firing order of an 8-cylinder V piston.

- a. 1-6-8-4-3-2-1-5
- b. 1-8-7-3-6-5-4-2
- c. All of the above
- d. None of the above

46. What is the mechanical efficiency of an engine having 8-hp brake horsepower and 10-hp indicated horsepower?

- a. 40 %
- b. 60 %
- c. 80 %
- d. None of the above

47. If the compression ratio is 1:17, the clearance volume for 100 cm³ piston displacement volume is ____.

- a. 6.25 cc
- b. 17.00 cc
- c. 20.00 cc
- d. None of the above

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48. The engine mechanical efficiency.

- a. Brake hp over indicated hp
- b. Rated hp over indicated hp
- c. Friction hp over flywheel hp
- d. None of the above

49. The number system indicating the ignition quality of diesel fuel.

- a. Octane rating
- b. Cetane rating
- c. Fuel index
- d. None of the above

50. An engine system that reduces the temperature of the cylinder.

- a. Ignition
- b. Carburetion
- c. Cooling
- d. None of the above

51. A 10-hp engine is equivalent to ____.

- a. 7.46 kW
- b. 76.4 watt
- c. 0.746 kW
- d. None of the above

52. A substance that decreases the effect of friction and is commonly obtained from petroleum products.

- a. Lubricant
- b. Gear oil
- c. Grease
- d. All of the above

53. High viscosity engine oil is desirable to use because ____.

- a. its viscosity changes easily with a change in temperature
- b. its viscosity does not change easily with the change in temperature
- c. its viscosity changes even without a change in temperature
- d. None of the above

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54. Gear oil is used for ____.

- a. engine
- b. transmission box
- c. hydraulic system
- d. All of the above

55. The mean effective pressure of a turbo-charged engine is ____.

- a. 300 psi
- b. 200 psi
- c. 400 psi
- d. None of the above

56. An engine valve that allows the entrance of fuel.

- a. Exhaust valve
- b. Intake valve
- c. Check valve
- d. None of the above

57. The standard throttling adjustment for engines.

- a. 1/3
- b. 1/2
- c. 3/4
- d. None of the above

58. The mean effective pressure of conventional engines is ____.

- a. 120 psi
- b. 200 psi
- c. 50 psi
- d. None of the above

59. The recommended additive to use to increase the resistance of engine oil and prevent the formation of certain acidic product that may cause corrosion of certain type of bearings.

- a. Anticorrosive
- b. Antioxidants
- c. Dispersants
- d. All of the above

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60. The term used to describe the resistance of oil to flow.

- a. Viscosity
- b. Cetane number
- c. Volume weight
- d. None of the above

61. An additive that provides a protective film on bearing surfaces thus preventing any acid to cause corrosion on metal parts.

- a. Anticorrosive
- b. Antioxidants
- c. Dispersants
- d. All of the above

62. If a 4-cylinder, 2-stroke-cycle internal combustion engine has 600 explosion per minute, what is the engine rpm? (Problem 5)

- a. 125 rpm
- b. 150 rpm
- c. 200 rpm
- d. None of the above

63. Considering that the engine (item 62) has 4" piston diameter and 5" piston stroke, what is its indicated horsepower assuming a mean effective pressure of 102 psi.

- a. 9.7 Hp
- b. 12.3 Hp
- c. 18.5 Hp
- d. None of the above

64. What is the brake horsepower of the engine in Item 62 if the mechanical efficiency is 80 percent. (Problem 5)

- a. 8 hp
- b. 12 hp
- c. 16 hp
- d. None of the above

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65. An engine that entirely uses oil as lubricant.

- a. Four-stroke cycle
- b. Two-stroke cycle
- c. All of the above
- d. None of the above

66. One horsepower can generate heat energy equal to ____.

- a. 2455 BTU/hr
- b. 2545 BTU/hr
- c. 2445 BTU/hr
- d. None of the above

67. If a 4-cylinder, 4 stroke-cycle internal combustion engine has 600 explosion per minute, what is the engine rpm?

- a. 50 rpm
- b. 75 rpm
- c. 100 rpm
- d. None of the above

68. The engine speed is automatically regulated by ____.

- a. hand lever
- b. governor
- c. flywheel
- d. None of the above

69. What is the brake thermal efficiency of an engine that uses 0.6-lb of fuel per horsepower-hr. The fuel contains 20,000 BTU/lb.

- a. 21.2 %
- b. 34.5 %
- c. 46.8 %
- d. None of the above

70. The ignition timing for a diesel engine.

- a. 5-12 deg.
- b. 17-19 deg
- c. 20-26 deg
- d. None of the above

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71. The indicated power of an engine is 69 horsepower and the brake horsepower is 54. What is the mechanical or engine efficiency.

- a. 69 %
- ☒ b. 78 %
- c. 59%
- d. None of the above

72. The ignition timing for gasoline engines.

- a. 1-5 deg.
- ☒ b. 5-10 deg.
- c. 10-15 deg.
- d. None of the above

73. If a 2-stroke, 4-cylinder engine is running at 2000 rpm, its power stroke per minute is ____.

- a. 4000
- ☒ b. 8000
- c. 2000
- d. None of the above

74. The number of pistons for a 6-cylinder 4-stroke gasoline engine.

- a. 3
- ☒ b. 6
- c. 9
- d. None of the above

75. The fuel for a compression ignition engine.

- a. Gasoline
- b. Kerosene
- c. LPG
- ☒ d. None of the above

76. An engine oil having high flash and fire point indicates ____.

- ☒ a. high boiling point
- b. high evaporation rate
- c. high heating value
- d. All of the above

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77. A four-stroke-cycle diesel engine, with three cylinders, has 89-mm cylinder bore and 130-mm stroke. Its compression ratio is 16:1 and it is running at 2000 rpm speed. What is the piston displacement of the engine per cylinder?

- ☒ a. 809 cc
- b. 1320 cc
- c. 650 cc
- d. None of the above

78. What is the total power stroke per minute of the piston in Item 77 above?

- a. 2000
- ☒ b. 3000
- c. 500
- d. None of the above

79. Referring to Items 77 & 78, what is the total displacement volume per minute of the engine? (Problem 17)

- ☒ a. 2427 lpm
- b. 2891 lpm
- c. 3014 lpm
- d. None of the above

80. Based on the above items, what is the piston speed of the engine?

- ☒ a. 520 m/min
- b. 672 m/min
- c. 725 m/min
- d. None of the above

81. Still referring to Item 77, what is the stroke-to-bore ratio of the engine.

- a. 1.00
- b. 1.32
- ☒ c. 1.46
- d. None of the above

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82. A four-cylinder gasoline engine has 20-cm cylinder diameter and 40-cm stroke. The cylinders run at a speed of 1500 rpm. The engine is a four-stroke cycle with 8 kg/cm² mean effective pressure. What is the area, in cm², of the piston?

- ☒ a. 314 cm²
- b. 284 cm²
- c. 167 cm²
- d. None of the above

83. What is the length of stroke of the engine piston, in meter, in Item 82 above?

- ☒ a. 0.40
- b. 0.20
- c. 0.8
- d. None of the above

84. What is the power stroke of the engine per cylinder in Item 82?

- a. 50 power stroke per minute
- ☒ b. 750 power stroke per minute
- c. 100 power stroke per minute
- d. None of the above

85. What is the total power stroke of the engine in Item 82?

- a. 150
- ☒ b. 3000
- c. 75
- d. None of the above

86. Still referring to Item 82, what is the indicated horsepower of the engine?

- a. 460 hp
- ☒ b. 660 hp
- c. 760 hp
- d. None of the above

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87. If the mechanical efficiency of the engine in Item 82 is 80 %, what is its brake horsepower?

- ☒ a. 530 Hp
- b. 350 Hp
- c. 650 Hp
- d. None of the above

88. What is the stroke-to-bore ratio of the engine in Item 82?

- ☒ a. 2.0
- b. 1.6
- c. 2.4
- d. None of the above

89. An engine running without load is in ____.

- a. optimum running condition
- b. low running condition
- ☒ c. idle condition
- d. None of the above
- e. All of the above

90. The engine speed, in revolution per minute, specified by the manufacturer.

- a. Rated horsepower
- b. Engine speed
- ☒ c. Rated engine speed
- d. All of the above

91. The thermal efficiency of an internal combustion engine.

- a. 0 to 10 %
- b. 10 to 20 %
- ☒ c. 20 to 30 %
- d. None of the above

92. Water flushing of the radiator should be done ____.

- a. every month
- ☒ b. every 3 months or 10,000 km
- c. every 6 months or 24,000 km
- d. All of the above

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93. An additive that cause carbon particles in the engine to remain in suspension in oil.

- a. Anticorrosive
- b. Antioxidant
- c. Dispersant
- d. All of the above

94. The burning of the portion of fuel in the combustion chamber of an engine at a rate faster than the desired.

- a. Knocking
- b. Detonation
- c. Misfiring
- d. All of the above

95. A device used to measure the quantity of oil in the crankcase.

- a. Dipstick
- b. Oil gauge
- c. Oil stick
- d. None of the above

96. A device that provides a rich mixture of fuel during high load and lean mixture when under load.

- a. Carburetor
- b. Economizer
- c. Fuel regulator
- d. All of the above

97. There are two engines, one is rated 1 hp and the other is 1 ps. Which of the following statements is true?

- a. One-hp engine is stronger than 1-ps engine.
- b. One-ps engine is stronger than 1- hp engine.
- c. The two engines have equal strength.
- d. None of the above.

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98. The act of analyzing, testing and measuring the engine to remedy the cause of trouble.

- a. Servicing
- b. Tune-up
- c. Trouble shooting
- d. All of the above

99. Japanese engines are usually rated in terms of ____.

- a. horsepower
- b. pferde starke
- c. watts
- d. All of the above

100. The type of gasoline without ethylene bromide or that with 87 octane rating, which sometimes causes knocking of engine.

- a. Premium
- b. Leaded
- c. Unleaded
- d. All of the above

101. The lubrication system of an engine ____.

- a. reduces its wear and tear
- b. acts as seal between the piston and the cylinder wall
- c. acts as complement of the cooling system
- d. All of the above

102. The term used to indicate leakage or loss of pressure as leakage of compression between the piston and the cylinder of a gasoline engine.

- a. Blow-by
- b. Head loss
- c. Back pressure
- d. None of the above

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GOD BLESS!!!

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